HP-200 Hydraulic Forming Press — Maintenance Manual

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Section 1: Routine Inspection & Lubrication Schedules

Regular preventative maintenance ensures structural alignment and hydraulic integrity of the HP-200 press. All maintenance activities must be performed by certified industrial maintenance technicians.

Daily Inspection Items:

1. Inspect all flexible hydraulic hoses and swivel joints for sweating, abrasions, or fluid weeping.
2. Check main system pressure gauge and pilot pressure gauge for steady readings.
3. Check oil reservoir temperature gauge and verify chiller water inlet valve is open.
4. Confirm emergency stop circuit and dual-channel safety gate interlocks function properly.

Weekly and Monthly Inspection Schedules:

Guide Column Lubrication: Grease the four chrome-plated vertical guide columns weekly using NLGI Grade 2 lithium complex grease. Pump 15 grams of grease into each of the four lower and upper bronze bushing grease fittings. Wipe off contaminated excess grease.

Platen Alignment and Parallelism: Check upper and lower platen parallelism monthly using a dial indicator across four bed corners. Total runout must not exceed ± 0.05 mm across the entire 800 mm x 800 mm platen area. Re-shim guide gibs if clearance exceeds 0.08 mm.

Accumulator Nitrogen Pre-charge: Check nitrogen pre-charge pressure on the bladder accumulator every 3 months. Nominal pre-charge is 90 bar (N2 only; never use compressed air or oxygen).

Section 2: Filter Maintenance & Hydraulic Troubleshooting

Filter Replacement Intervals:

The high-pressure inline hydraulic oil filter must be replaced every 500 operating hours, or immediately if the visual differential pressure indicator extends into the red band (indicating $\Delta P > 3.5$ bar). Always inspect hydraulic hoses and manifold connections for leaks at every filter change. Reservoir hydraulic fluid must be replaced every 2,000 operating hours or annually after oil laboratory spectrometry testing.

Troubleshooting Repeated Low Hydraulic Pressure:

When repeated low hydraulic pressure alarms or cycle sluggishness occurs, collect the following systematic observations before replacing hydraulic components:

1. Record pump discharge pressure under deadhead dead-end condition and active forming load.
2. Measure case drain leakage rate from the variable displacement axial piston pump: normal leakage is under 1.5 L/min; case drain flow exceeding 4.5 L/min at 200 bar indicates severe cylinder barrel or valve plate wear requiring pump rebuild.
3. Inspect proportional pressure relief valve cartridge seat for erosion, particle contamination, or broken bias spring.
4. Check differential pressure indicator on the main pressure line filter to rule out severe filter element clogging.
5. Verify suction strainer in reservoir is not cavitating due to clogged 100-mesh wire cloth.

Tie Rod Pre-Tensioning:

Inspect and torque tie rod tension nuts every 1,000 operating hours. Tighten in cross-pattern to 1,200 Nm using a hydraulic torque wrench.